

COMPSCI 5100

ML & AI for Data Science

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Unit 2: Classification

Classification = Automatically label data

Automatic speech recognition (ASR)



welcome

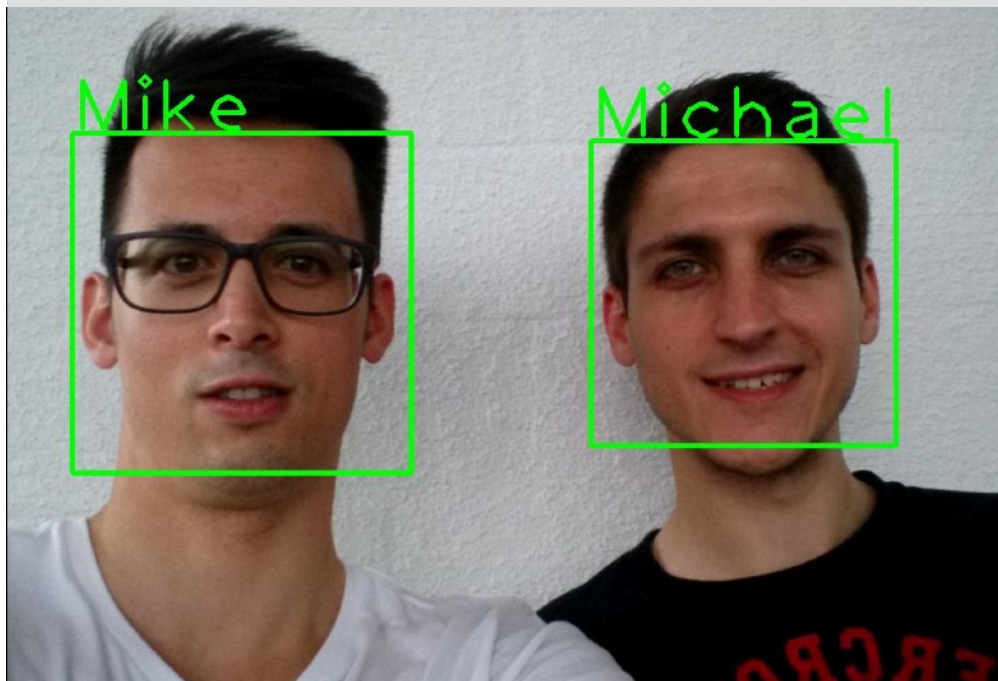


hello



not speech

Face recognition



Sentiment analysis



My experience
so far has been
fantastic!

POSITIVE



The product is
ok I guess

NEUTRAL



Your support team is
useless

NEGATIVE

Learning Objectives

- To be able to formulate a problem to a classification task
- To understand the components of a classifier
- To be able to evaluate a classifier's performance

Classification: Part I

ML paradigms

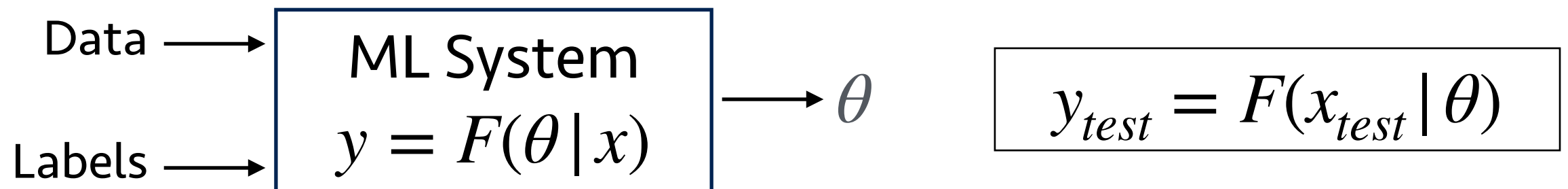
[AI $\neq$ ML, but in today's world ML and AI are often synonymous]

Machine Learning paradigms

- **Supervised**
 - need labeled training data
- **Semi-supervised**
 - some labeled training data
- **Unsupervised/ Self-supervised**
 - no labeled training data needed

Supervised learning paradigm

- **Observe** a set of examples:
 - **Training data** (measurements of any kind) = x
 - **Labels** = y
- **Model** the relationship between data and labels
- **Predict** the label for a new data (**test data**)



Classification vs. Regression

Classification

- **Lables** = y is **discrete**
- For example, y identifies an image as: *dog, cat, mouse* (**class**)
- As the number of classes increase, difficulty of a classification task increases

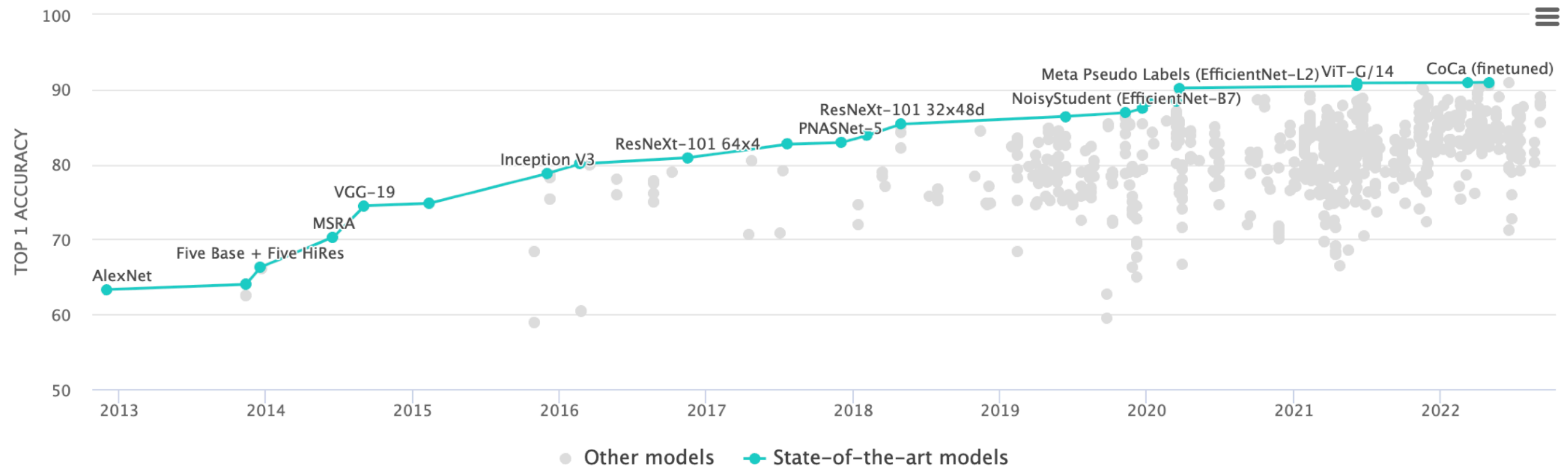
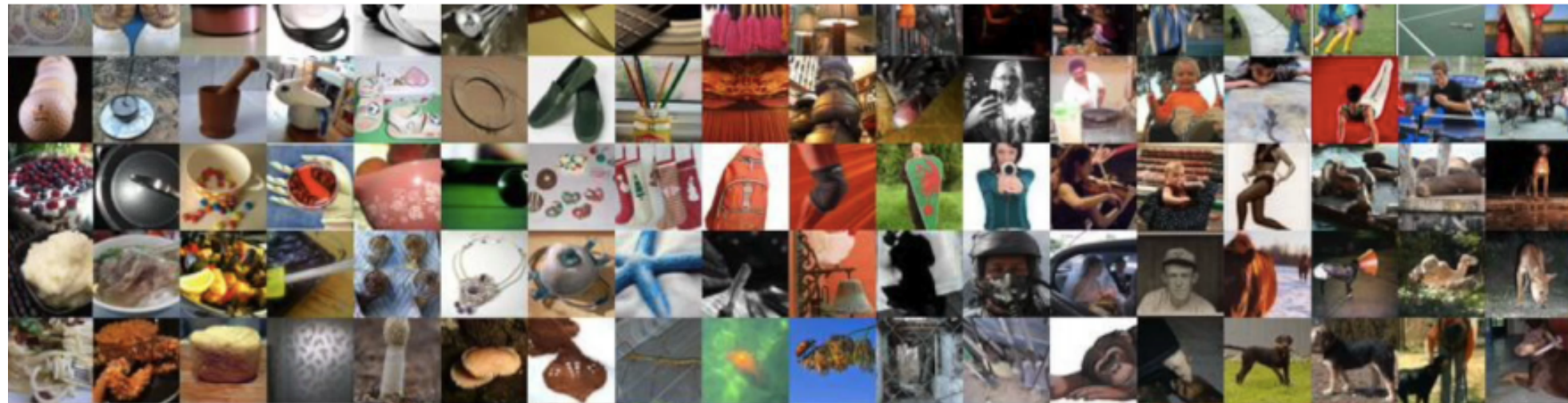
Regression

- **Lables** = y is **continuous**
- For example, y predicts the price of houses in Glasgow

Can we transform a regression task to a classification task?

Success stories

Imagenet challenge (>1000 classes)

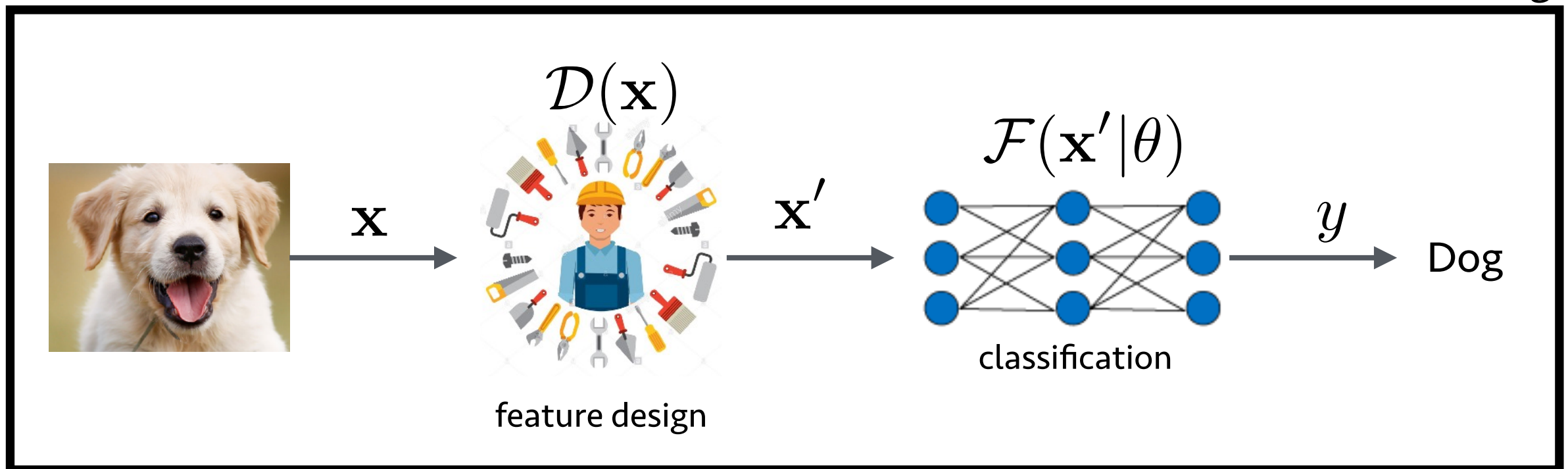


Components of a classification system

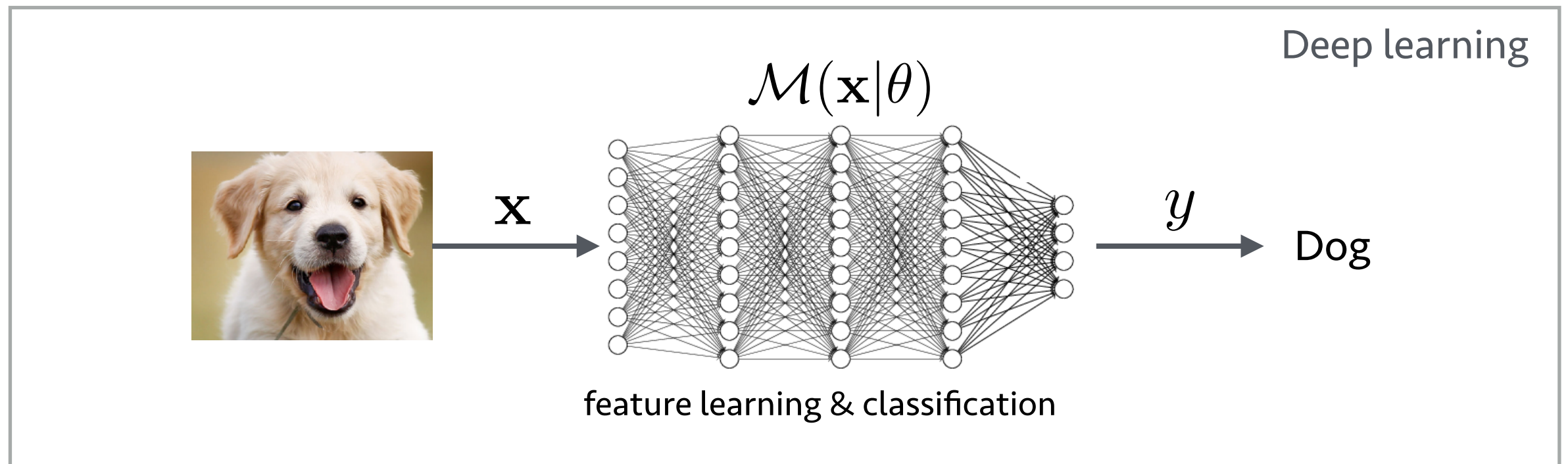
1. Labeled data for training
2. Features (hand crafted for traditional ML, learned in DL)
3. Model
4. Evaluation metrics

Components of a classification system

Traditional machine learning

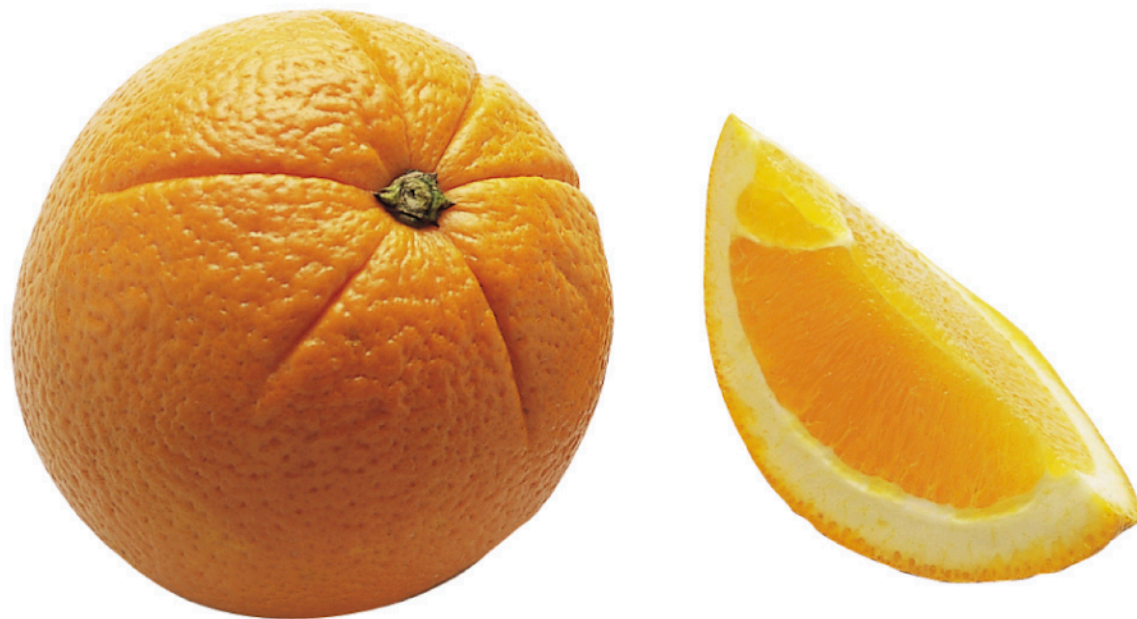


Deep learning



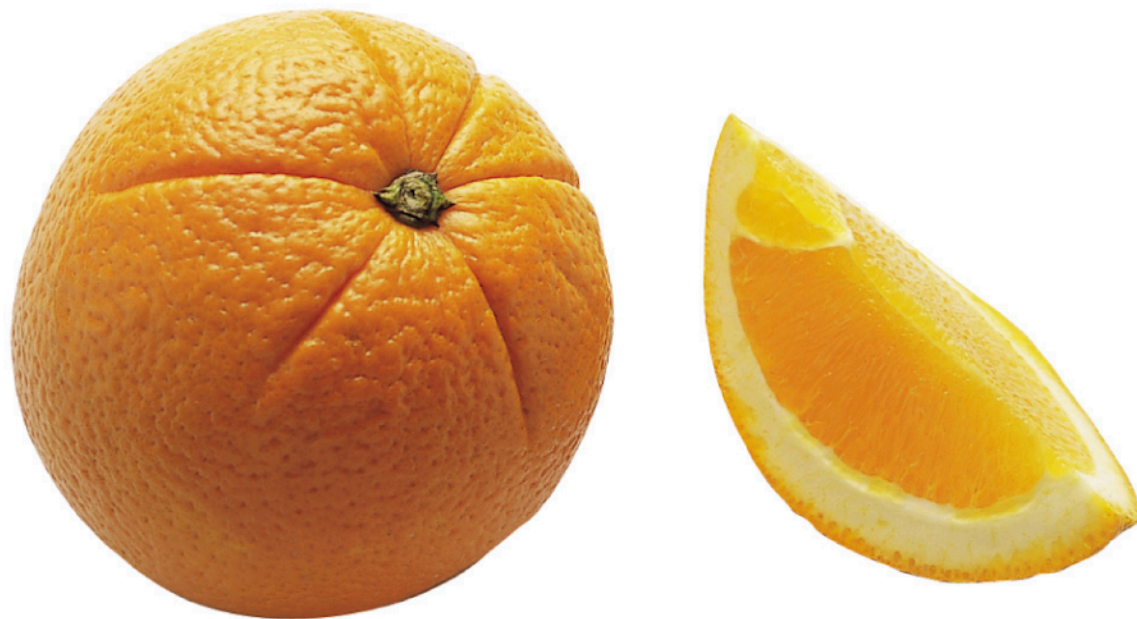
Example

Task: Build a classification system to identify oranges and lemons
(not necessarily using images)



Example

Task: Build a classification system to identify oranges and lemons

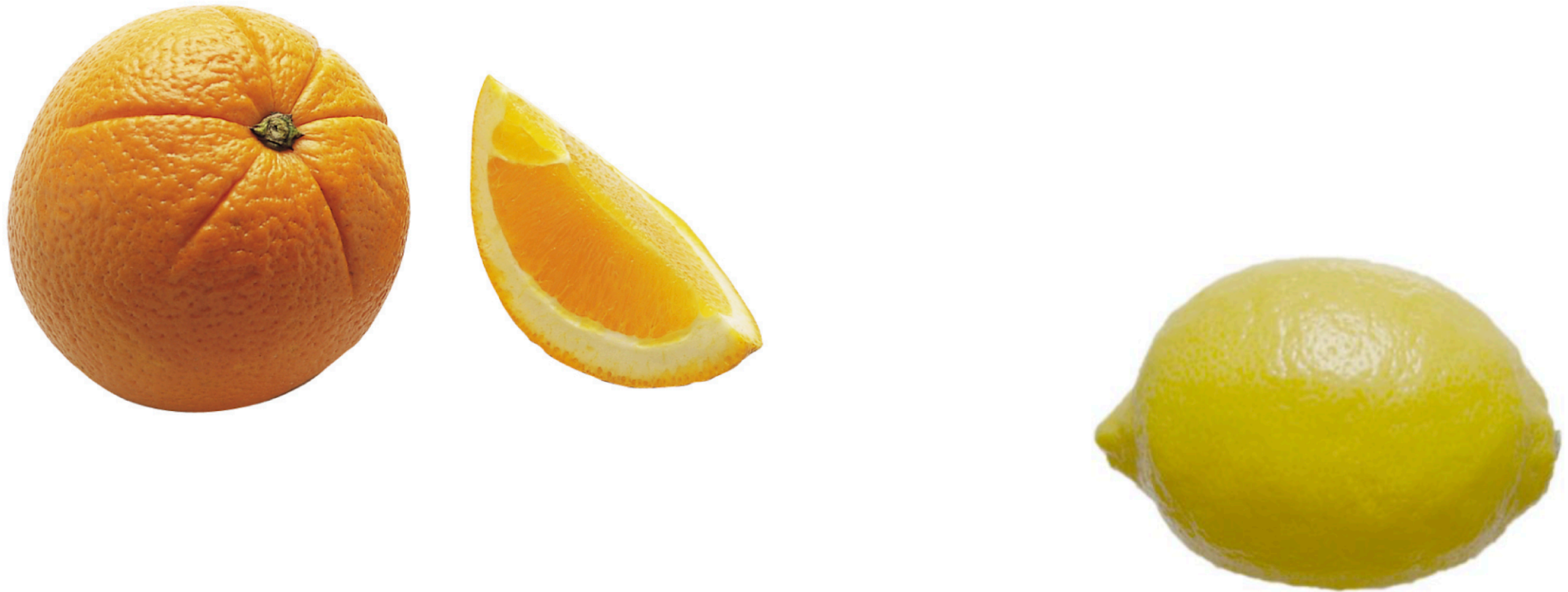


Step 1: Decide which attributes can you measure

- weight
- shape
- colour (images)

Example

Task: Build a classification system to identify oranges and lemons



Step 2: Collect data to create training set

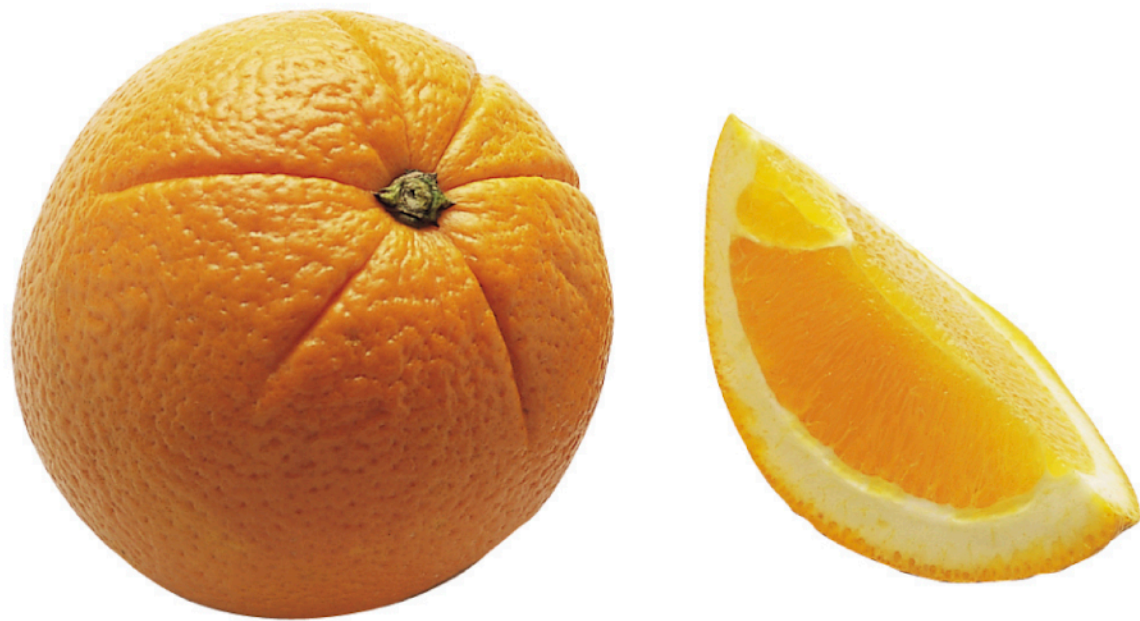
(Ideal) Training data

- **ML models can not recognise data they do not see**
- Need a (large) set of training data
- Balanced samples across classes
- Not noisy (accurate measurement)
- Varied examples for each class



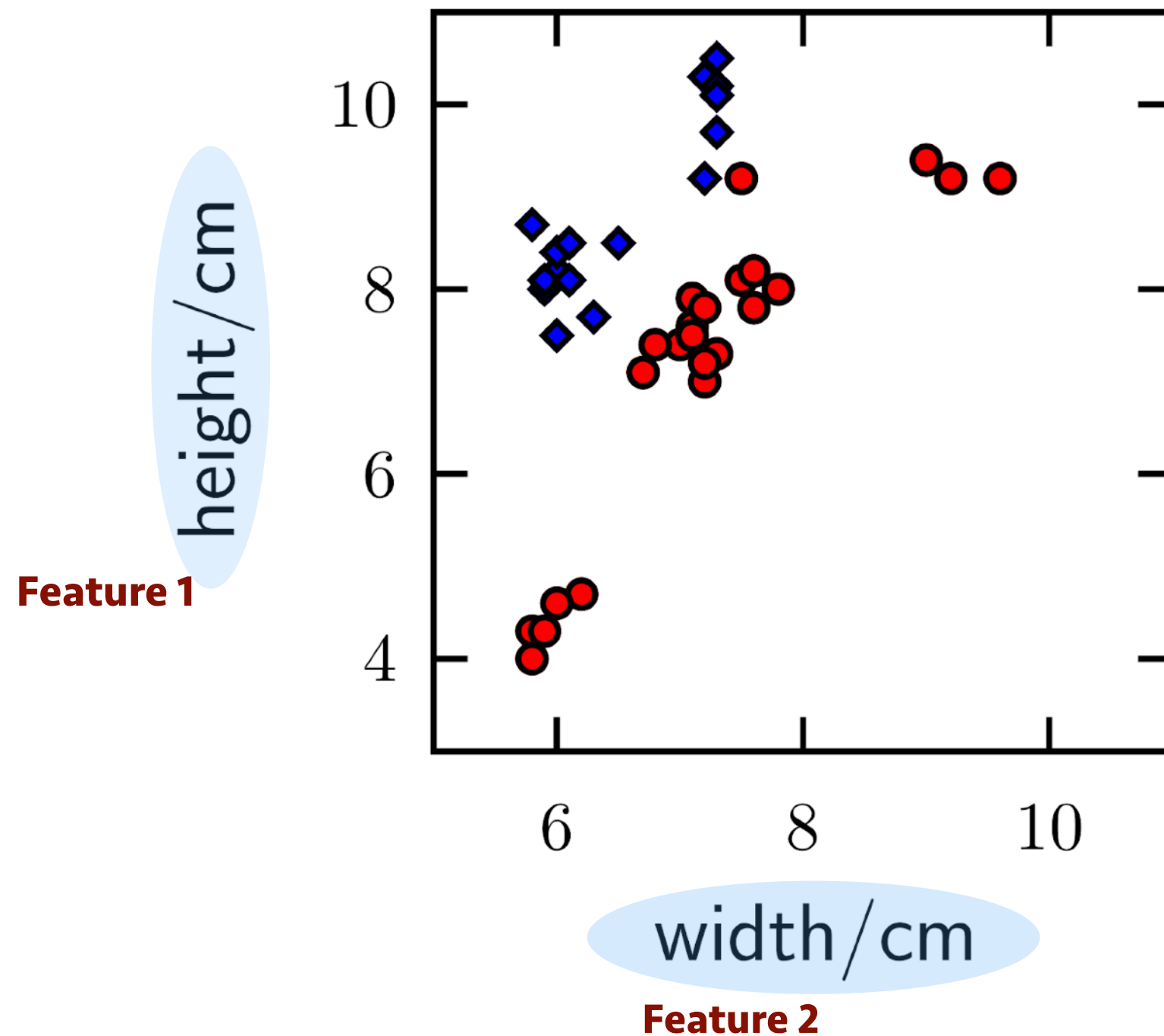
Example

Task: Build a classification system to identify oranges and lemons

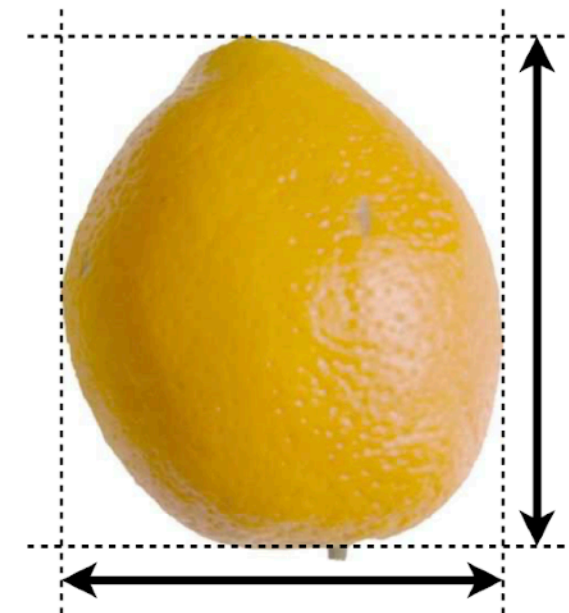


Step 3: Extract features

Features



Oranges: ●
Lemons: ◆



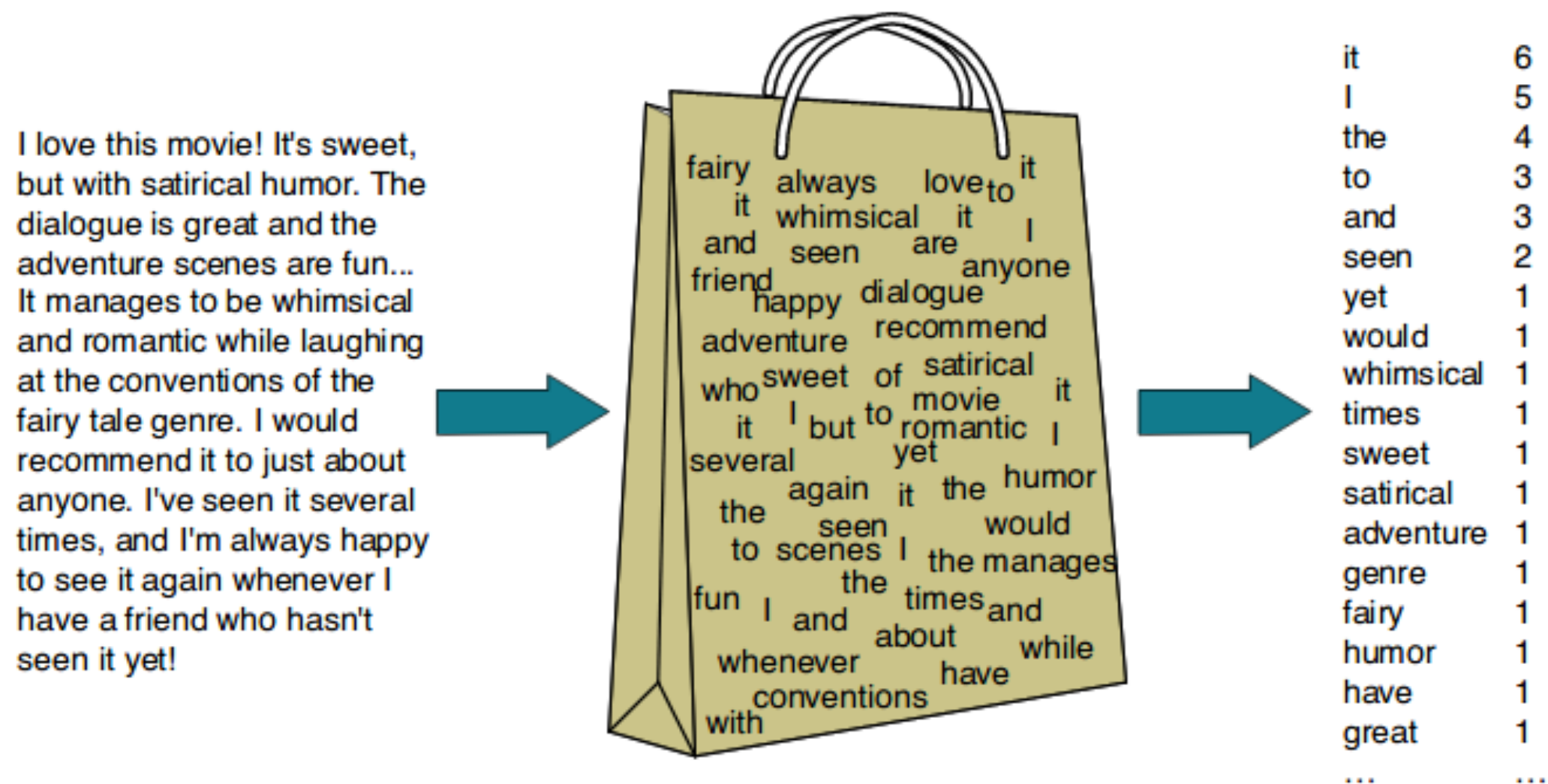
Features

Features = Representation of Data

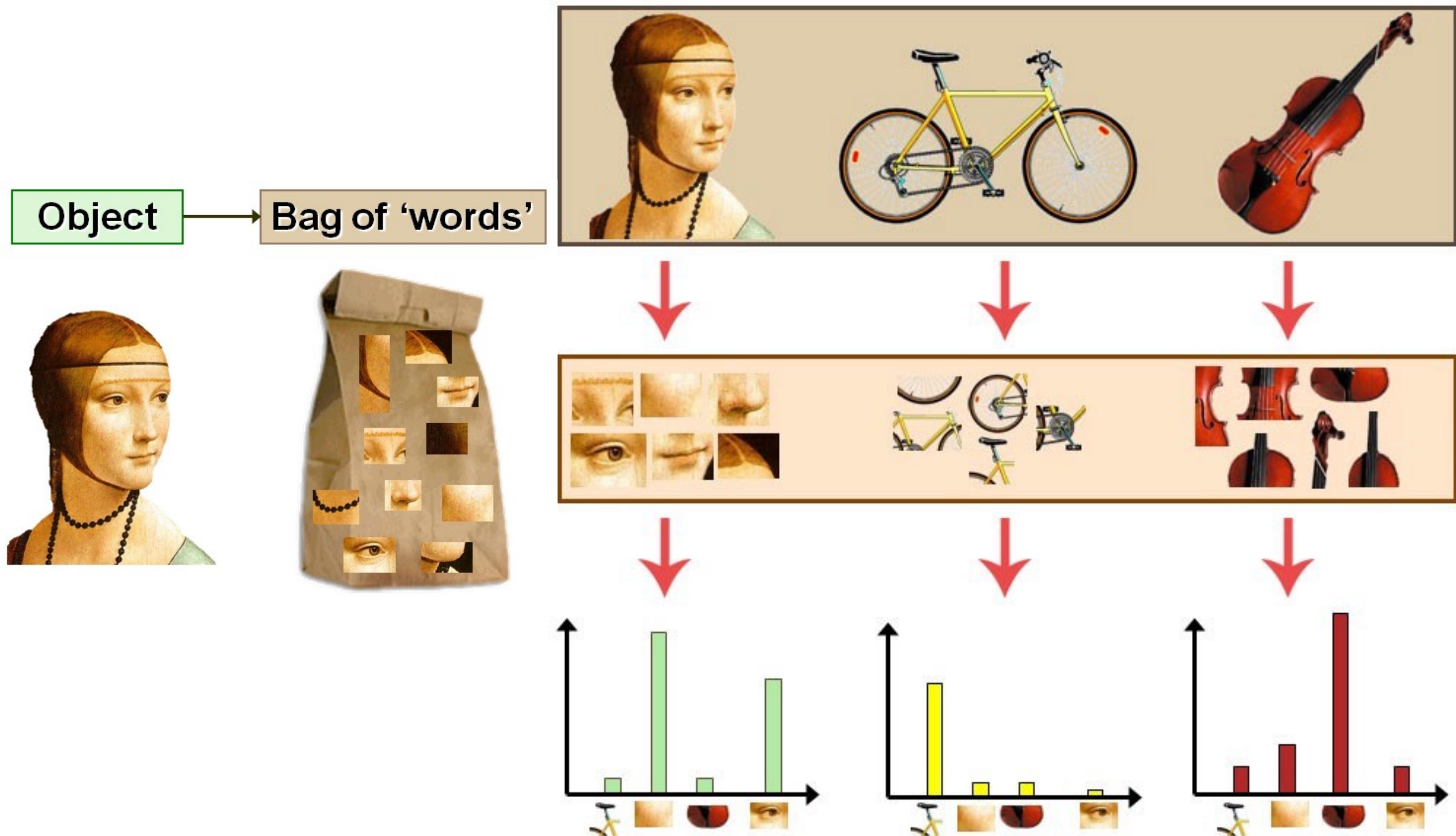
- Features have high impact on model performance
- For classification, features should be discriminative
- Depends on data type (modality): image, text ...
 - Needs domain knowledge
- Depends on application
- Usually difficult to identify the 'right' set of features
 - Rely on prior work, Trial and error
 - Systematic 'selection' of features

Features example: Bag of Words (BoW)

Features depend on data type



Features example



Features

Features						Label
Name	Egg-laying	Scales	Poisonous	Cold-blooded	Number legs	Reptile
Cobra	1	1	1	1	0	1
Rattlesnake	1	1	1	1	0	1
Boa constrictor	0	1	0	1	0	1
Chicken	1	1	0	1	2	0
Guppy	0	1	0	0	0	0
Dart frog	1	0	1	0	4	0
Zebra	0	0	0	0	4	0
Python	1	1	0	1	0	1
Alligator	1	1	0	1	4	1

Features

5 Features (Feature dimension = 5)

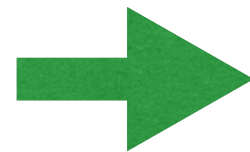
Name	Egg-laying	Scales	Poisonous	Cold-blooded	Number legs	Feature Vector
Cobra	1	1	1	1	0	→ [1 1 1 1 0]
Rattlesnake	1	1	1	1	0	→ [1 1 1 1 0]
Boa constrictor	0	1	0	1	0	.
Chicken	1	1	0	1	2	.
Guppy	0	1	0	0	0	
Dart frog	1	0	1	0	4	
Zebra	0	0	0	0	4	
Python	1	1	0	1	0	.
Alligator	1	1	0	1	4	→ [1 1 0 1 4]

Feature selection

- More features → high dimensional features
 - high dimension → higher model complexity (curse of dimensionality) → may or may not lead to high performance
- **Not all features are equally important**
- Reduce feature space by throwing out unimportant features. Find a 'parsimonious' model.
 - Occam's razor: *simplest explanation that accounts for the data is the best*

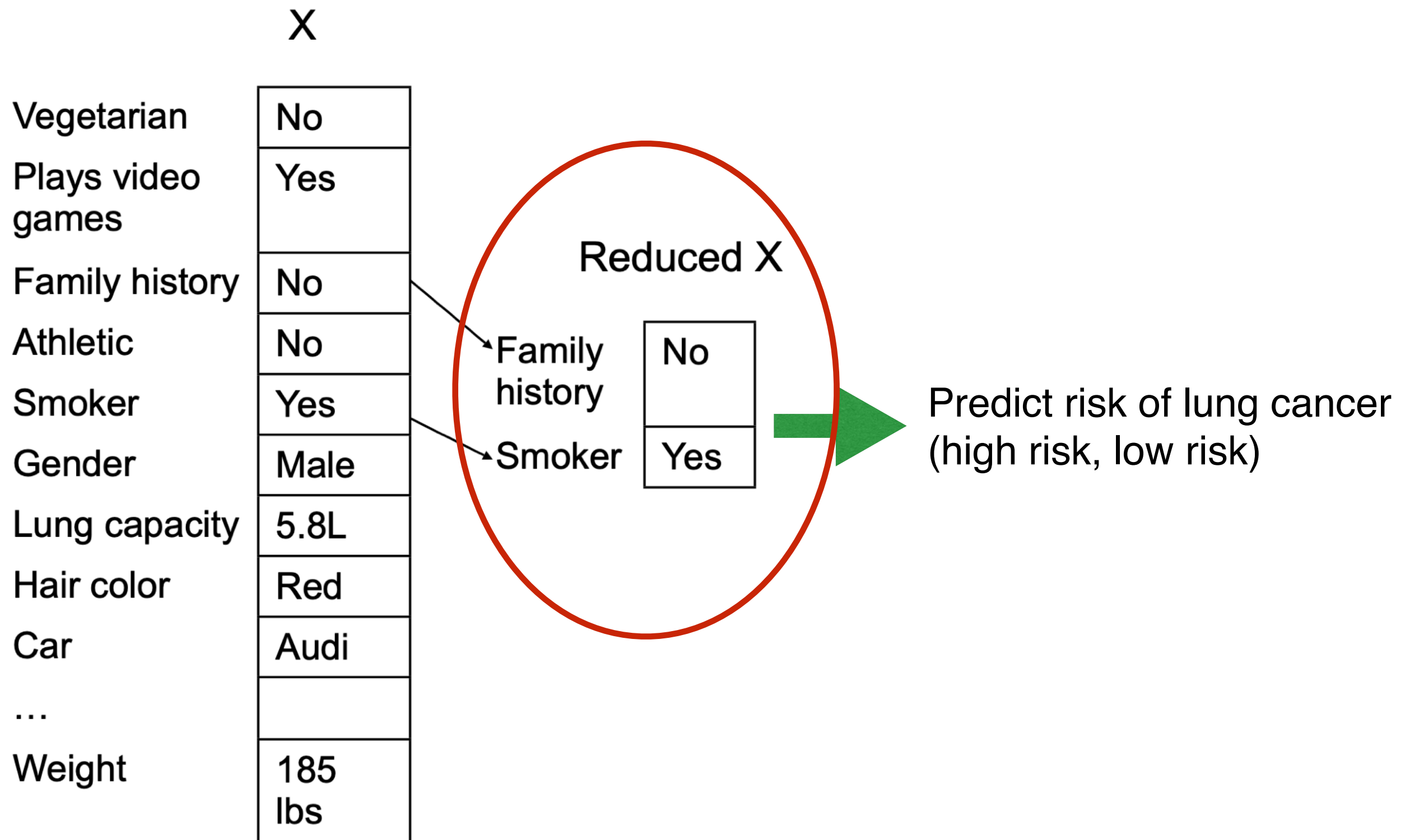
What is feature selection

Vegetarian	No
Plays video games	Yes
Family history	No
Athletic	No
Smoker	Yes
Gender	Male
Lung capacity	5.8L
Hair color	Red
Car	Audi
...	
Weight	185 lbs



Predict risk of lung cancer
(high risk, low risk)

What is feature selection



Feature selection

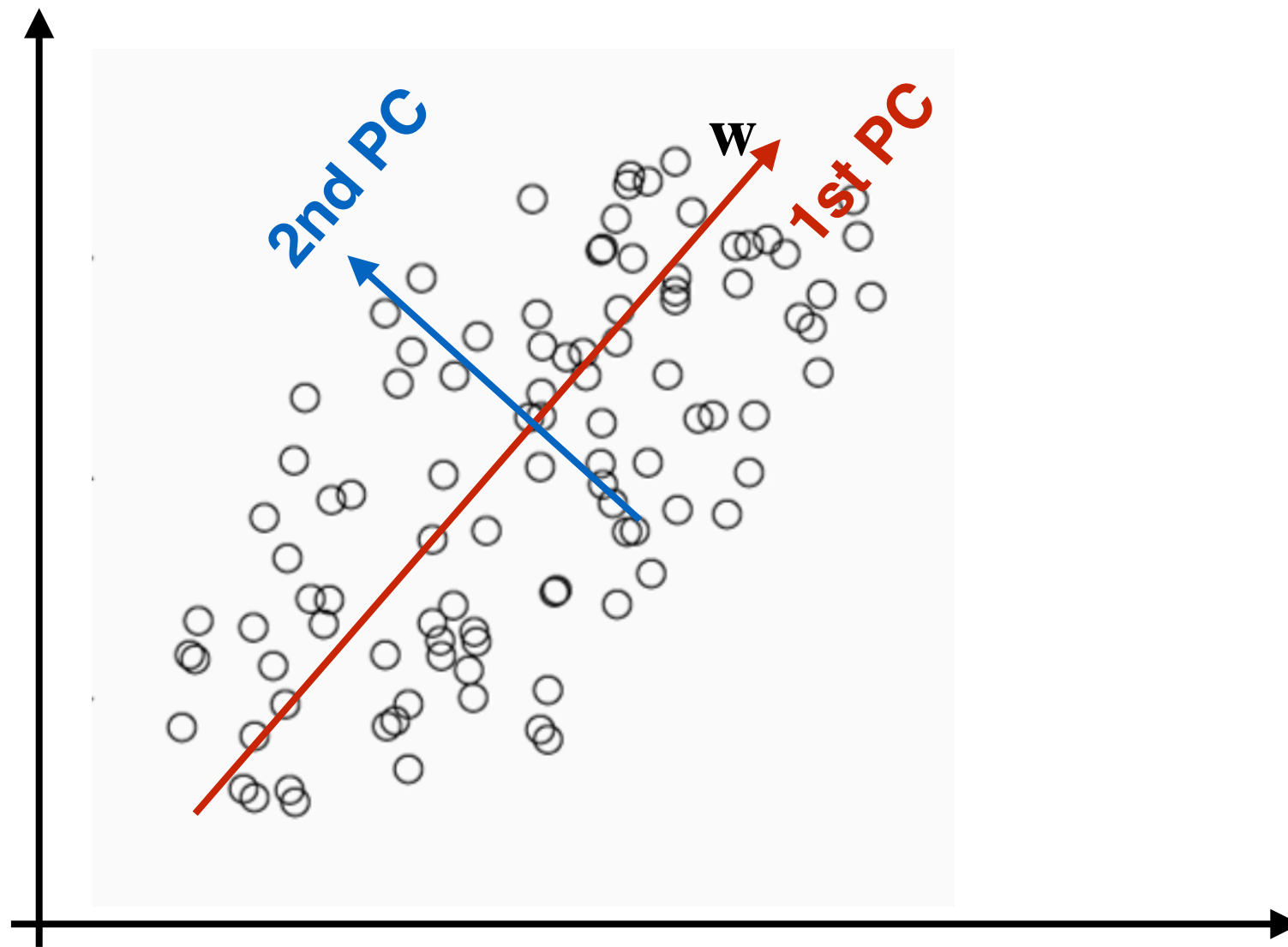
- How to go from $X \rightarrow$ Reduced X ?
- **Dimensionality reduction**
 - Principal Component Analysis (PCA) tells us which features account for the highest **variance** in data.
- Several other methods for feature selection
 - More on this during Case Studies in Week 9/10

PCA

- Consider a Feature matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ where n is the number of training samples, p is the dimension of features

PCA

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PCA

- Consider a Feature matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ where n is the number of training samples, p is the dimension of features
- **First PC** $\mathbf{w}$: Find projection of $\mathbf{X}$ on $\mathbf{w}$ such that

$$\begin{aligned}\mathbf{w}^* &= \operatorname{argmax}_{\|\mathbf{w}\|=1} \|\mathbf{X}\mathbf{w}\|^2 \\ &= \operatorname{argmax}_{\|\mathbf{w}\|=1} \|\mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w}\|^2 \\ &= \operatorname{argmax}_{\mathbf{w}} \left(\frac{\mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w}}{\mathbf{w}^T \mathbf{w}} \right) \longleftarrow \text{Rayleigh Quotient}\end{aligned}$$

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- Max value of Rayleigh quotient is given by the largest eigenvalue of $\mathbf{C} = \mathbf{X}^T \mathbf{X}$. Therefore $\mathbf{w}^*$ is the corresponding eigenvector.

PCA

- PCs are eigenvectors of $\mathbf{C}$, their importance being weighted by their corresponding eigenvalues.
- $\mathbf{C} = \mathbf{X}^T \mathbf{X}$ is called **Covariance matrix** (scatter matrix)
- We just need to diagonalize $\mathbf{C} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^{-1}$, where $\mathbf{\Lambda}$ contains eigenvalues and $\mathbf{V}$ contains the eigenvectors (orthonormal).
 - Why is $\mathbf{C} \in \mathbb{R}^{p \times p}$ more convenient?
- Columns of $\mathbf{V}$ are the PCs. They are **orthogonal**.
- In practice, we standardise $\mathbf{X}$ (mean 0, unit norm).
- PCA is just a **linear transformation** (change of coordinates) - does not change any information in the data.

Dimensionality reduction via PCA

- Input: Feature matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ where n is the number of training samples, m is the dimension of features
- Output: Reduced feature matrix $\mathbf{X}_d \in \mathbb{R}^{n \times d}$ where $d < p$
- Compute $\mathbf{C} = \mathbf{X}^T \mathbf{X}$ and $\mathbf{C} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^{-1}$
- Keep only the **largest** d eigen values in $\mathbf{\Lambda}$ and their corresponding eigenvectors
 - $\mathbf{\Lambda} \rightarrow \mathbf{\Lambda}_d$ and $\mathbf{V} \rightarrow \mathbf{V}_d$ where $\mathbf{V}_d \in \mathbb{R}^{p \times d}$
- Project: $\mathbf{X}_d = \mathbf{X} \mathbf{V}_d$

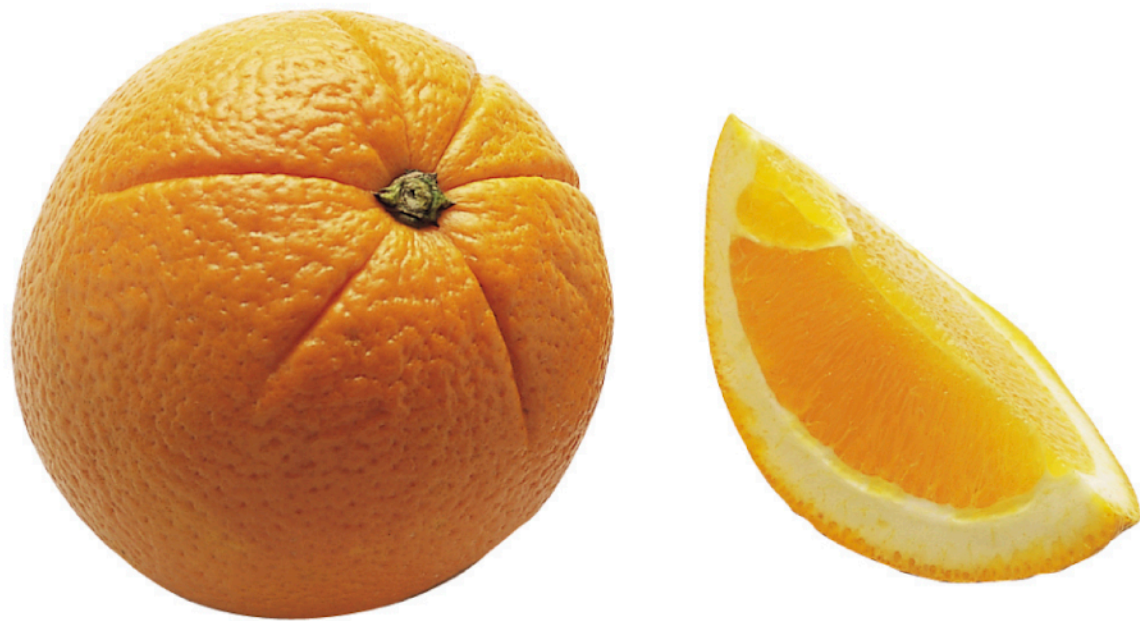
Linear algebra refresher

Recommended

- <https://stanford.edu/~shervine/teaching/cs-229/refresher-algebra-calculus>
- <https://cs229.stanford.edu/section/cs229-linalg.pdf>

Example

Task: Build a classification system to identify oranges and lemons



Step 4: Build ML Model (classifier)

Summary: Part I

- Learning paradigms
 - supervised, unsupervised, semi-supervised
- Components of a Classification System
 - Data, features, model, evaluation metrics
- Data and Features (feature selection) discussed
- Model and Evaluation metrics coming up